A Data-Integrated Multi-Omics Approach to Study Cancer Related Fatigue

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Background and Purpose

Cancer-related fatigue (CRF) is a highly prevalent symptom reported that occurs during and persists following cancer treatment. CRF has significant negative outcomes on patients' mood and ability to function. Research is needed to identify mechanisms and develop prediction models for CRF severity (Kober & Yom, PMID 34028000). In a sample of oncology patients (n=1343) undergoing chemotherapy (CTX), the first aim of our NCI R37 (CA233774) is to evaluate underlying mechanisms for CRF, by comparing patients with high versus low levels of CRF in terms of gene expression (GE), genetic, and epigenetic changes (i.e., "omics"). Given that no treatments are available for fatigue, this approach may identify potential therapeutic interventions. Our second aim is to develop a risk prediction model that could assist with the identification of patients at greatest risk for high levels of CRF. The approach that is being used in this R37 will be applied to future investigations of other patient reported outcomes (e.g., cognitive impairments, Oppegaard and Harris et al. PMID 34388477).

Integrated Data Analysis

CRF is a complex phenotype, which is the result of environmental (e.g., demographic, clinical, social) and genetic factors. By integrating data from multiple sources, we will have increased power to identify and better interpret the omics-phenotype relationships relative to an analysis that uses only a single source of omics data. We will use this strategy to obtain a more complete biological model of the common and distinct molecular factors associated with the severity of morning and evening CRF. In systems genomics, a multi-staged data integrated analysis can be used, which follows a step-wise or hierarchical approach. In this study, we will perform a multi-staged analysis to identify potential mechanisms for CRF (Harris et al. PMID 33827270).

First, we will use a subset of ~60% of the patients to identify associations between functional gene products (i.e., GE) and the severity of morning and evening CRF. Using this list of functional candidate genes, we will evaluate the relative contributions of heritability (i.e., genetics) and environment (i.e., epigenetics) to the CRF phenotypes in the remaining subset of patients (Figure 2). Identifying these candidate genes and pathways will improve our understanding of the mechanism(s) of CRF and provide new therapeutic targets.

This multi-staged data-integrated multi-omics approach will dramatically reduce the number of tests performed, and increase the power to detect significant and clinically meaningful common and distinct associations with morning and evening CRF. In addition, by performing the discovery DE analysis on a different set of patients than the genetic and epigenetic association analyses we limit the bias inherit in the potential correlations between GE and methylation and genetic variations. Here, we report on the current gene expression and epigenetic findings.

Prediction of CRF Severity

The use of predictive models that include phenotypic and molecular characteristics has led to improvements in risk profiling and treatment for a several cancers. We will use machine learning methods, that incorporate the rich demographic, clinical, and omic data, to develop a valid predictive model of CRF. This prediction modeling may determine not only which phenotypic characteristics are the most salient in predicting CRF severity, but the relative contribution of omic characteristics to this outcome.

Study Sample

Oncology outpatients (n=1346) who were receiving CTX for breast, lung, gynecological, or gastrointestinal cancers were recruited as part of the parent grant (NCI R01 CA134900). Patients beginning their 2nd or 3rd cycle of CTX were assessed over two complete CTX treatment cycles (i.e., 6 assessments) using the Lee Fatigue Scale (LFS). In the week prior to their 2nd or 3rd cycle of CTX (i.e., Assessment), patients were asked to rate the severity of CRF in the past week upon awakening (morning CRF) and 30 minutes prior to bedtime (evening CRF). Blood samples were drawn for the omics analyses. Differences in phenotypic characteristics between the groups were evaluated using the appropriate univariate tests. The characteristics that are most relevant to this grant are summarized below. Morning and evening CRF groups were created based on the LFS scores at Assessment 1 (CRF prior to the 2nd or 3rd cycle of) using clinically meaningful cutpoints for morning (i.e., >3.2) and evening (i.e., >5.6) fatigue.

Diurnal Variations in CRF

Morning Fatigue Sample. Compared to the low group (55.4%, n=724, mean morning LFS score=1.42, SD=0.96), patients in the high morning CRF group (44.6%, n=583, mean morning LFS score=5.24, SD=1.49) were significantly younger, had a higher body mass index (BMI), a lower functional status score, a higher level of comorbidity, a longer time since diagnosis, a greater number of prior cancer treatments, and a higher number of metastatic sites. In addition, compared to the low group, patients in the high morning CRF group were more likely to be female, more likely to live alone, less likely to be married/partnered, more likely to have child care responsibilities, less likely to be currently employed, more likely to have an annual household income of <$30,000, more likely to have depression, less likely to exercise on a regular basis, and more likely to be diagnosed with breast cancer.

Evening Fatigue Sample. Compared to the low group (50.2%, n=647, mean evening LFS score=3.64, SD=1.47), patients in the high morning CRF group (49.8%, n=642, mean evening LFS score=7.05, SD=1.12), were significantly younger, had a lower functional status score, and a higher level of comorbidity. In addition, compared to the low group, patients in the high evening CRF group were more likely to be female, to be White, to have child care responsibilities, to be employed, to have depression, to have high blood pressure, and to be diagnosed with breast cancer.

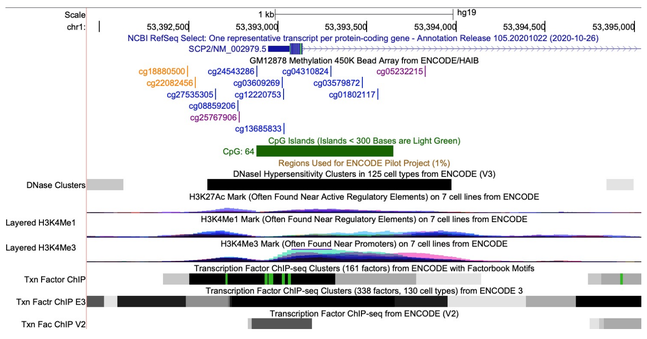

Figure 1. UCSC Genome browser (hg19). Putative regulatory region tracks are included. Differential methylation in patients with low versus high morning fatigue was identified at SCP2 eCpG cg01802117.

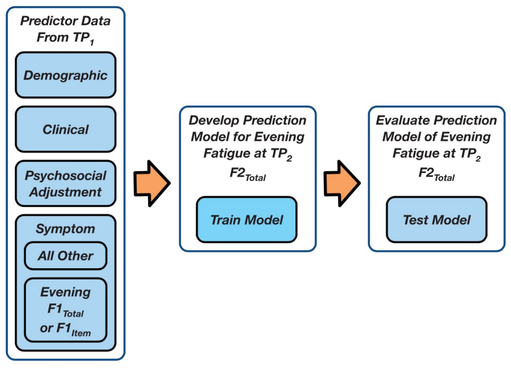

Figure 2. A depiction of the data collected at Time Point 1 (TP_1) used to develop the models to predict evening fatigue at Time Point 2 (TP_2). Evening fatigue at TP1 is characterized as either the total score ($F1_{Total}$) or scale items ($F1_{Item}$) of the Lee Fatigue Scale.

Stage 1. Expression Functional Candidates

Gene expression analyses were performed in two independent samples using RNA-sequencing (S1, n=349) and Illumina HumanHT-12 v4.0 microarray (S2, n=349) methodologies. Differential expression was quantified using generalized linear models separately for each sample (i.e., S1: edgeR, S2: limma). These analyses were adjusted for demographic and clinical characteristics that significantly differed between patients with low and high morning fatigue. In addition, the models included surrogate variables not associated with morning fatigue to adjust for variation due to unmeasured sources or unwanted artifacts using surrogate variable analysis. Only genes that had a common direction of expression (i.e., the same sign for the log fold-change) across the two samples were retained for the PIA. Fisher's combined probability test was used to determine significantly perturbed pathways between the two morning fatigue groups across both samples (Bonferroni corrected alpha=.05).

Evening CRF

High levels of evening CRF was reported by 52.1% of the patients in S1 and by 54.7% in S2. Across the two samples, 29 pathways were found to be perturbed between patients with low and high levels of evening CRF. Examples of the perturbed KEGG signaling pathways include: antigen processing and presentation, cytokine-cytokine receptor interaction, chemokine, and intestinal immune network for IgA production. This is the first study to identify perturbations in inflammatory pathways are involved in evening fatigue in patients undergoing CTX across multiple diagnoses.

Morning CRF

High levels of morning fatigue was reported by 53.9% of the patients in S1 and by 46.7% in S2. Across both samples, 39 pathways were found to be perturbed between patients with low and high levels of morning fatigue. Examples of the perturbed KEGG signaling pathways include: PPAR, relaxin, oxytocin, serotonergic synapse, cytokine-cytokine receptor interaction and chemokine. This is the first study to identify perturbations in neuroendocrine and inflammatory pathways that are associated with morning fatigue among patients receiving CTX.

Stage 2. Epigenetics of Functional Candidates

In this pilot study, we evaluate for differences in regulatory regions of genes in the most significantly perturbed neurotransmitter signaling pathway (i.e., PPAR), exploratory tests for differentially methylated probes (DMPs) between the morning fatigue groups. Methylation levels were measured using the Illumina HumanMethylation450 microarray. A linear model was fit using limma that included significant demographic and clinical characteristics associated with morning fatigue group membership in the total S2. Compositions were estimated for blood cell types (i.e., CD8T, CD4T, neutrophile, NK, B, and monocyte) and tested for an association with fatigue group membership. Ten genes were represented by twenty-four CpG loci in promoter regions associated with gene expression changes (eCpG) identified in two independent datasets of human blood cells. The type I error threshold for this exploratory study of candidate DMPs was set at 0.05 and not adjusted for multiple comparisons.

In this exploratory analysis of a subset of patients diagnosed with breast cancer in S2 (n=116) for which DNA methylation data were available, 52.6% reported high morning fatigue. Cell type estimates were not associated with fatigue group membership. The final regression model included the seven demographic and clinical characteristics identified in the regression analysis from the total S2. Morning fatigue group was significantly associated with methylation levels of one of the eCpG loci for the Sterol carrier protein (SCP2) gene (cg01802117; p-value = 0.048; Figure 1). We were unable to identify previous studies associating methylation with morning CRF levels. However, genetic variation and gene expression levels of SCP2 are associated with narcolepsy. This pilot study is the first to evaluate eCpGs of genes in perturbed pathways. Methylation data were recently collected using the Illumina EPIC microarray for the majority of the study participants (n=940). Future studies will evaluate for DM of genes in the perturbed pathways identified in Stage 1 for morning and evening CRF.

Prediction Modeling of CRF Severity

The purpose of this study is to predict the severity of morning and evening fatigue in the week following the administration of CTX using machine learning approaches. Outpatients (N=1217) completed questionnaires one week prior to and one week following administration of CTX. Separate prediction models for fatigue severity were created using clinical, symptom, and psychosocial adjustment characteristics and either total fatigue scores or individual fatigue item scores (Figure 2). This approach has the potential to assist clinicians to identify high-risk patients and potentially modifiable risk factors (e.g., high levels of anxiety), as well as novel therapeutic targets. Future studies will include 'omics data (i.e., genetic, transcriptomic, and epigenomic) predictors. These findings will guide the development of future clinical (e.g., risk reduction strategies, pre-emptive treatment of co-occurring symptoms) and preclinical (e.g., novel genomic markers) studies to decrease the severity of CRF.

Evening CRF severity one week following administration of CTX

Prediction models were created using multivariable linear regression, RPART, random forest (RF), and support vector machine. RF models provided the best fit across all models. For the RF model using individual LFS item scores, two of the 13 individual LFS items (i.e., "worn out", "exhausted") were the strongest predictors. An important clinical finding is the ability of the RF model using single items from evening LFS scores obtained prior to chemotherapy to predict the severity of evening fatigue post-chemotherapy as accurately as the model using total evening LFS scores. This finding suggests that clinicians can ask patients to rate (0-10) their level of feeling "worn out" or "exhausted" prior to chemotherapy to estimate their evening fatigue in the week following chemotherapy. The use of these descriptive single items may facilitate the assessment of evening fatigue in a busy oncology clinic (Kober et al. PMID 34249477).

Morning CRF severity one week following administration of CTX

Prediction models were created using multivariable linear regression, RPART, RF, support vector machine, LASSO, and ElasticNet (EN). EN models provided the best fit across all models. For the EN model using individual LFS item scores, Karnofsky Performance Score, two of the 13 individual LFS items (i.e., "worn out", "exhausted"), quality of sleep, and excessive daytime sleepiness were the strongest predictors. This finding highlights the predictive value of the two LFS item terms and supports the hypothesis that morning and evening fatigue are similar but distinct symptoms.

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